

## Case I: Continuous RV's

With 2 continuous r.v.'s,  $X$  and  $Y$ , we will still have

$$P(X=x, Y=y) = 0 \text{ for any } x, y.$$

So as before, we must describe the joint distribution of continuous r.v.'s through a density.

But instead of obtaining the prob. of a (1-dimensional) interval, we use the density to compute the prob. of (2-dimensional) rectangles.

Def. (Joint PDF): If  $X, Y$  are two continuous r.v.'s, their joint pdf,  $f(x, y)$ , must satisfy:

- $f(x, y) \geq 0$ , for all  $x, y$

- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

- For any two intervals  $[a_1, b_1], [a_2, b_2]$ :

$$P(X \in [a_1, b_1], Y \in [a_2, b_2]) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x, y) dy dx$$

The following def's (marginal, conditional) are similar to the discrete case, only replacing sums (e.g. over  $y$ ) by integrals.

Def. (Marginal PDF): The marginal pdf of  $X$  is equal to

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy.$$

Likewise,  $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$

Note: The marginal pdf of  $X$  will satisfy, for any  $[a, b]$ :

$$\int_a^b f_X(x) dx = P(X \in [a, b]) = P(X \in [a, b], Y \in (-\infty, \infty)) = \int_a^b \int_{-\infty}^{\infty} f(x, y) dy dx$$

Def. (Conditional PDF): The conditional pdf of  $X$  given  $Y=y$  is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

Note: Still, these density functions can't be interpreted as " $IP(X=x, \text{ and } Y=y)$ ", or " $IP(X=x|Y=y)$ ".

If  $X, Y$  are continuous, it is still the case that  $IP(X=x, \text{ and } Y=y) = 0$  for all  $x, y \dots$

Hence, the conditional pdf doesn't have as "nice" of an interpretation as the conditional pmf.

Now, we can turn to the notion of independence for continuous joint r.v.'s.

Def. (Indep. R.V.'s - Cts): If  $X, Y$  have joint density  $f(x, y)$ , then we say  $X$  and  $Y$  are independent if

$$f(x, y) = f_X(x) f_Y(y), \text{ for all } x, y.$$

Observation: If we have some two intervals  $[a_1, b_1]$ , and  $[a_2, b_2]$ , and  $X, Y$  are independent, then:

$$\begin{aligned} IP(X \in [a_1, b_1], Y \in [a_2, b_2]) &= \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x, y) dy dx \\ &= \int_{a_1}^{b_1} \int_{a_2}^{b_2} \underline{f_X(x)} f_Y(y) dy dx && \left. \begin{array}{l} X, Y \text{ indep.} \\ f_X(x) \text{ is constant over } y \end{array} \right\} \\ &= \int_{a_1}^{b_1} f_X(x) \left[ \int_{a_2}^{b_2} f_Y(y) dy \right] dx \\ &= \int_{a_1}^{b_1} f_X(x) \underbrace{\left[ \int_{a_2}^{b_2} f_Y(y) dy \right]}_{= IP(Y \in [a_2, b_2]) \text{ by def'n, and is constant in } x} dx \\ &= IP(Y \in [a_2, b_2]) \cdot \underbrace{\int_{a_1}^{b_1} f_X(x) dx}_{= IP(X \in [a_1, b_1])} \end{aligned}$$

$$\Rightarrow P(X \in [a_1, b_1], Y \in [a_2, b_2]) = P(Y \in [a_2, b_2]) \cdot P(X \in [a_1, b_1])$$

I.e., the two events  $\{X \in [a_1, b_1]\}$  and  $\{Y \in [a_2, b_2]\}$  are independent, for any choices of  $[a_1, b_1]$  and  $[a_2, b_2]$

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Again, we may have more than two continuous random variables, and want to describe their distribution.

Def. (Joint/Marginal PDF,  $n$  continuous r.v.'s): For continuous r.v.'s  $X_1, \dots, X_n$ ,

Their joint pdf,  $f(x_1, \dots, x_n)$ , satisfies:

$$P(X_1 \in [a_1, b_1], \dots, X_n \in [a_n, b_n]) = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} f(x_1, x_2, \dots, x_n) dx_n \dots dx_1$$

for any intervals  $[a_1, b_1], \dots, [a_n, b_n]$ .

The marginal pdf,  $f_{X_k}(x_k)$ , for any  $k=1, \dots, n$ , is:

$$f_{X_k}(x_k) = \underbrace{\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty}}_{n-1 \text{ integrals}} f(x_1, \dots, x_n) \underbrace{dx_n \dots dx_{k+1} dx_{k-1} \dots dx_1}_{\text{over } n-1 \text{ variables - all except } x_k}$$

Note: The marginal pdf of  $X_k$  is obtained by "integrating out" the possible values of the other  $X_1, \dots, X_{k-1}, X_{k+1}, \dots, X_n$ .

It satisfies  $\int_a^b f_{X_k}(x_k) dx_k = P(X_k \in [a, b]) = P(X_1 \in (-\infty, \infty) \dots X_k \in [a, b] \dots X_n \in (-\infty, \infty))$

Def. (Mutually Indep. R.V.'s - Cts): The random variables  $X_1, \dots, X_n$  are mutually independent if their joint density satisfies:

$$f(x_1, x_2, \dots, x_n) = f_{X_1}(x_1) \cdot f_{X_2}(x_2) \cdot \dots \cdot f_{X_n}(x_n)$$

for all  $x_1, \dots, x_n$ .

Observation: If  $X_1, \dots, X_n$  are mutually independent, then for any intervals  $[a_1, b_1], [a_2, b_2], \dots, [a_n, b_n]$ , we have

$$P(X_1 \in [a_1, b_1], \dots, X_n \in [a_n, b_n]) = P(X_1 \in [a_1, b_1]) \cdot \dots \cdot P(X_n \in [a_n, b_n])$$

$\Rightarrow$  The events  $\{X_1 \in [a_1, b_1]\}, \dots, \{X_n \in [a_n, b_n]\}$  are mutually independent, for any intervals  $[a_1, b_1], \dots, [a_n, b_n]$ .

Note: There is also a definition for pairwise independent random variables. We will not define it in this class, simply because it will not be used.

So, if we write "let  $X_1, X_2, \dots, X_n$  be independent r.v.'s", it can safely be assumed that  $X_1, X_2, \dots, X_n$  are mutually independent.

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Example: Consider the joint density of two continuous r.v.'s  $X, Y$ , given by:  $f(x, y) = \frac{3}{2} \cdot e^{-3y}$ , for  $x \in [0, 2], y \geq 0$

Q1: Is this a valid joint pdf?

A1: +  $f(x, y) \geq 0$ , so this is ok.

+ Check:  $\iint f(x, y) dx dy = 1$

$$\int_0^{\infty} \left[ \int_0^2 \frac{3}{2} e^{-3y} dx \right] dy$$

↑ This remains constant over  $x \in [0, 2]$ ,  
so we can pull it out

$$= \int_0^{\infty} \left[ \frac{3}{2} e^{-3y} \int_0^2 1 \cdot dx \right] dy$$

= 2 - 0 = 2

$$= \int_0^{\infty} \frac{3}{2} e^{-3y} \cdot 2 \cdot dy = \int_0^{\infty} 3e^{-3y} dy$$

$$= \left[ 3 \cdot \frac{1}{2} e^{-3y} \right]_0^{\infty} = \left( \lim_{y \rightarrow \infty} -e^{-3y} \right) - (-e^{-3 \cdot 0})$$

$$= 0 - (-1) = \underline{1} \quad \checkmark$$

$\Rightarrow f(x,y)$  integrates to 1!

$\leadsto$  Yes, this is a valid joint density.

Q2: What are the marginal dist's of  $X, Y$ ?

A2: Marginal of  $X$ :

$$f_X(x) = \int_0^{\infty} f(x,y) dy$$

$$= \int_0^{\infty} \frac{3}{2} \cdot e^{-3y} dy$$

$$= \frac{1}{2} \cdot \int_0^{\infty} 3 \cdot e^{-3y} dy = \frac{1}{2}, \text{ for all } x \in [0,2].$$

$\Rightarrow$  The marginal pdf of  $X$  is  $f_X(x) = \frac{1}{2}, \forall x \in [0,2]$   
 $\leadsto X \sim \text{Unif}(0,2) !$

Marginal of  $Y$ :

$$f_Y(y) = \int_0^2 f(x,y) dx$$

$$= \int_0^2 \frac{3}{2} \cdot e^{-3y} dx$$

Constant in  $x$ , so pull it out

$$= \frac{3}{2} \cdot e^{-3y} \int_0^2 1 \cdot dx$$

$\int_0^2 1 \cdot dx = 2$

$$= \frac{3}{2} \cdot e^{-3y} \cdot \cancel{2} = \underline{3 \cdot e^{-3y}}$$

$\Rightarrow$  The marginal pdf of  $Y$  is  $f_Y(y) = 3 \cdot e^{-3y}$ , for  $y \geq 0$ .  
 $\leadsto Y \sim \text{Exp}(3) !$

Q3: Compute the conditional pdf of  $X$ , given  $Y=y$ .

A3: 
$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$$
$$= \frac{\frac{3}{2} e^{-3y}}{3 \cdot e^{-3y}} = \frac{1}{2}, \text{ for any } x \in [0,2] \text{ and } y \geq 0$$

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Q4: Are  $X$  and  $Y$  independent?

A4: Yes! 2 ways we might see this:

i) The conditional pdf of  $X$  (given  $Y=y$ ) is the same as the marginal density of  $X$ .

$$f_{X|Y}(x|y) = \frac{1}{2}, \text{ for } x \in [0,2], y \geq 0$$

$$f_X(x) = \frac{1}{2}, \text{ for } x \in [0,2]$$

$\Rightarrow$  Observing the value of  $Y=y$  doesn't change the distribution of  $X$ , regardless of what  $y \geq 0$  is.

ii) Using the def'n of independence,

$$\left. \begin{aligned} f(x,y) &= \frac{3}{2} e^{-3y} \\ f_X(x) f_Y(y) &= \left(\frac{1}{2}\right) \cdot 3e^{-3y} \end{aligned} \right\} \text{ for } x \in [0,2], y \geq 0$$

$\Rightarrow$  Since  $f(x,y) = f_X(x) f_Y(y)$  for every  $x, y$ ,  
then  $X, Y$  are independent r.v.'s.

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## Expectations of Jointly Distributed R.V.'s:

Let us have some jointly distributed r.v.'s,  $X_1, X_2, \dots, X_n$

- ⊛ Notice that we can compute  $E[X_i]$  by:
- i) finding the marginal pdf (or pmf) of  $X_i$
  - ii) computing  $\int_{-\infty}^{\infty} f_{X_i}(x_i) x_i dx_i$  (or  $\sum x_i p_{X_i}(x_i)$ ) as usual.

What if we are interested in  $E[g(X_1, \dots, X_n)]$ , where  $g$  depends on all of  $X_1, \dots, X_n$ ?

Def. (Expectation - Joint R.V.'s): If  $X_1, \dots, X_n$  are joint r.v.'s and  $g(x_1, \dots, x_n)$  is real-valued, then:

$$E[g(X_1, \dots, X_n)] = \begin{cases} \sum_{x_1} \sum_{x_2} \dots \sum_{x_n} g(x_1, x_2, \dots, x_n) p(x_1, x_2, \dots, x_n), & \text{if } X_1, \dots, X_n \text{ discrete} \\ \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} g(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_n \dots dx_2, & \text{if } X_1, \dots, X_n \text{ continuous} \end{cases}$$

Note: We will not be looking at cases where e.g.  $X_1$  is discrete but  $X_2$  is continuous. Either i) all  $X_1, \dots, X_n$  are discrete, or ii) all of  $X_1, \dots, X_n$  are continuous.

These expectations can be very challenging to compute sometimes. However, there are some cases that are easy!

(e.g., ⊛ above for finding  $E[X_i]$ )

Prop. (Linearity of Expectation - Joint R.V.'s): If  $X_1, \dots, X_n$  are any jointly distributed r.v.'s (either discrete or continuous), then:

$$E[X_1 + X_2 + \dots + X_n] = E[X_1] + \dots + E[X_n]$$

Proof: Left to the bold student confident with multivariable calculus...

This is really useful, since it actually doesn't matter whether the r.v.'s  $X_1, \dots, X_n$  are independent.

However, if they are independent, there is another nice setting.

Prop. (Product of Indep. R.V.'s): If  $X_1, \dots, X_n$  are independent,

then:  $E[X_1 \cdot \dots \cdot X_n] = E[X_1] \cdot E[X_2] \cdot \dots \cdot E[X_n]$

Proof: For  $n=2$ , cts. r.v.'s

$$E[X_1 \cdot X_2] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 \cdot f(x_1, x_2) dx_2 dx_1$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{[x_1 \cdot f_{X_1}(x_1)]}_{\text{constant over } x_2} \cdot [x_2 \cdot f_{X_2}(x_2)] dx_2 dx_1$$

By indep. of  $x_1, x_2$

constant over  $x_2$ , so we pull it out.

$$= \int_{-\infty}^{\infty} x_1 \cdot f_{X_1}(x_1) \left[ \int_{-\infty}^{\infty} x_2 \cdot f_{X_2}(x_2) dx_2 \right] dx_1$$

$= E[X_2]$ , and is constant over  $x_1$

$$= E[X_2] \cdot \underbrace{\left[ \int_{-\infty}^{\infty} x_1 \cdot f_{X_1}(x_1) dx_1 \right]}_{= E[X_1]} \quad \square$$

From this result, we get to one of the most important properties of variance.



Prop. (Variance of Linear Comb.): If  $X_1, \dots, X_n$  are independent r.v.'s,

then: 
$$\text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n),$$

Proof: For  $n=2$ , and  $X_1, X_2$  cts. r.v.'s

$$\text{Var}(X_1 + X_2) = E[(X_1 + X_2 - E[X_1 + X_2])^2]$$

$$= E[(X_1 - E[X_1]) + (X_2 - E[X_2])^2]$$

$$= E[(X_1 - E[X_1])^2 + 2(X_1 - E[X_1])(X_2 - E[X_2]) + (X_2 - E[X_2])^2]$$

$$= E[(X_1 - E[X_1])^2] + E[(X_2 - E[X_2])^2]$$

$$+ 2 \cdot E[(X_1 - E[X_1])(X_2 - E[X_2])]$$

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2 \cdot E[X_1 X_2 - X_1 E[X_2] - X_2 E[X_1] + E[X_1] E[X_2]]$$

$$= E[X_1 X_2] - E[X_1] \cdot E[X_2] - E[X_2] \cdot E[X_1] + E[X_1] \cdot E[X_2]$$

$$= 0$$

Cancel terms after using  $E[X_1 X_2] = E[X_1] \cdot E[X_2]$  by indep.